

1-B:

This question relates to the **law of diminishing marginal utility**, which states that as a consumer consumes additional units of the same good, the **extra satisfaction (marginal utility)** derived from each additional unit will decline. Buying a second pair of socks (provided that they are identical) will provide additional utility, but that extra utility is likely **less** than what the consumer gained from the first pair. Thus, marginal utility decreases. As a result, key B, becomes correct.

A: Marginal productivity relates to **labor input** and not consumption. It's a production-side concept, so not relevant here.

C: Again, marginal productivity isn't affected by buying consumer goods.

D: This is the opposite of what diminishing marginal utility suggests. The second identical unit yields *less* utility, not more.

2-B:

An **indifference curve** shows all the combinations of two goods that provide a consumer with the **same level of total satisfaction or utility**. It is derived from ordinal utility theory and is typically downward-sloping and convex to the origin, reflecting the consumer's willingness to substitute one good for another while maintaining equal satisfaction.

A: This is the definition of a **budget line**, not an indifference curve.

C: Again, this refers to budget constraints, not preferences.

D: Marginal utility changes along the indifference curve, but the indifference curve itself doesn't show the *rate at which Marginal utility changes as consumption changes*.

3-B:

The table shows the total amount consumers are willing to pay for different quantities of good X and the total external benefits that arise from the consumption of X.

quantity of good X (000 units)	consumers' willingness to pay (\$000)	total external benefits (\$000)
1	100	20
2	180	38
3	240	54
4	280	68
5	300	80

For an additional 1000 units

The Marginal WTP is \$20,000

The Marginal external benefits are \$12,000

MSB = Marginal Private Benefit + Marginal External Benefit

We are asked for the **marginal social benefit (MSB)** at 5000 units.

MSB = marginal private benefit + marginal external benefit.

First, calculate marginal willingness to pay (this represents the additional value the consumer is placing on that extra unit):

From 4k to 5k units:

- Change in willingness to pay = $300,000 - 280,000 = \$20,000$

Next, calculate marginal external benefits:

- Change in external benefit = $80,000 - 68,000 = \$12,000$

So, MSB = \$20,000 + \$12,000 = \$32,000

Hence, B is correct.

4-C:

Note: A production function shows a relationship between a firm's inputs and outputs.

A **production function** shows the **output** obtainable from different combinations of **inputs** (like labor and capital)-in the long-run. It underpins the analysis of **returns to scale**, which describes how output responds when all inputs are scaled up.

A and B: Both incorrectly list "demand" — the production function is **supply-side**.

D: "Supernormal profit" is a result, not what the production function directly measures — it arises from revenue and cost comparisons.

5-B:

A merger can lead to **economies of scale**, causing **average costs to fall**.

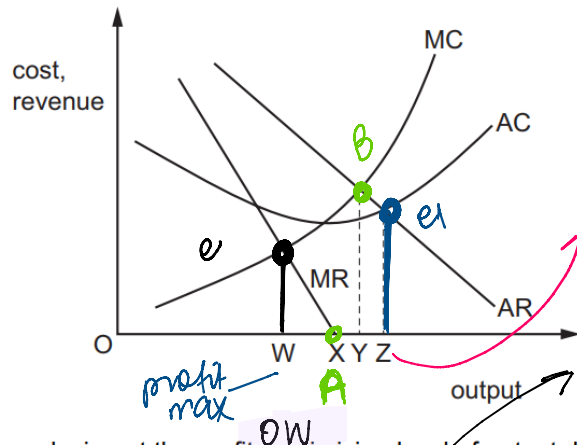
However, a larger firm with greater market share may face a **less elastic demand curve or a more "inelastic demand curve"**. This is because, a merger leads to a reduced competition in the market — that is, consumers may have fewer substitutes and thus be less responsive to price changes. Hence, the merged firm's demand becomes **more inelastic**.

This is why, key B, becomes correct.

A: "More elastic" is unlikely in a concentrated market; with less competition, consumers have fewer choices.

C and D: Rising average costs are inconsistent with the efficiency motivation behind merging in this case — unless there are diseconomies of scale, which aren't suggested here.

6 The diagram shows the costs and revenues of a firm operating in an imperfect market.



The firm is currently producing at the profit maximising level of output. It wishes to produce at the sales maximising level of output.

What would be the change in its output?

- A** OW to OX **B** OW to OY **C** OW to OZ **D** OY to OZ

- A) At O_X , $MR=0$, which represents the “sales revenue maximizing level of output”- since, it is not the sales maximizing level of output, key A, becomes incorrect.
- B) At O_Y , $MC=AR$ (price), which represents the allocatively efficient level of output- since, it is not the sales maximizing level of output, key A, becomes incorrect.
- D) O_Y is the allocatively efficient level of output (as explained above), instead of it being the profit maximizing level of output, and so, it is not where the firm is currently producing. The profit-maximizing level of output is where $MC=MR$, denoted through O_W . This is why, this becomes incorrect.

7-D:

Perfect competition is characterized by many small firms, homogenous products, and no barriers to entry. Since firms in perfect competition earn only **normal profits** in the long run and face perfectly elastic demand, they lack the profit margin or incentive to invest in **R&D**. Therefore, dynamic efficiency — which depends on innovation over time — is least likely.

A: Oligopolies can make super-normal profits in the long-run, and they often reinvest profits to stay ahead of rivals — classic examples include tech and pharma.

B: Monopolistic competition can allow some innovation due to brand differentiation (in the short-run)

C: Monopolies have supernormal profits in the long-run and may invest, though they might also be complacent and end up being X-inefficient— but less so than perfect competitors.

8-B:

The **principal-agent problem** arises when employees (agents) do not act in the best interests of the owners (principals). Linking worker pay to firm profitability directly aligns incentives.

This means that when the firm profits, the workers benefit as well apart from the shareholders. This motivates them to increase efficiency and performance.

A: Increasing salaries may reduce reliance on tips but doesn't necessarily link worker output to outcomes. (Incentives of the principal and the agent may not align directly)

C: Paying lawyers upfront to make him prepare a case well may not solve the principal-agent problem.

D: Permanent contracts offer job security, but again, don't create a performance-based incentive (Incentives of the principal and the agent may not align directly)

9-A:

Option A is correct because it's **forward vertical integration**. Company R (steel) is an input supplier to Company S (shipbuilding). If R takes over S, it moves upstream in the production chain, gaining control over its output market. This is why, key A, becomes correct.

10-D:

This is a classic **prisoner's dilemma**, a central concept in **game theory**. Firm X faces two options:

- **Act independently**, risking either a high payoff (\$900) or a low one (\$400), depending on its rival's unknown choice.
- **Cooperate**, earning a **guaranteed \$700**, which is less than the maximum but avoids the risk of the lowest outcome.

While **mutual cooperation** gives the best joint outcome (\$1400 total), **lack of trust** may push both to act selfishly, leading to a worse result.

This strategic uncertainty and interdependence are the core of the **prisoner's dilemma**. This explains why option D is correct.

- A) **Contestable markets** refer to markets with **low barriers to entry and exit** where firms behave competitively even if few firms exist because of the 'threat of potential competition.'

- B) **Kinked demand curve** represents **pricing in a non-collusive oligopoly**, where firms fear price wars if they lower prices and being undercut if they raise them. Therefore, prices “remain rigid at the kink”. This scenario given in the question is not about price rigidity.
- C) **Principal-agent problem** occurs when the **agent (e.g., manager)** has different incentives from the **principal (e.g., the shareholder)**. This situation involves **two firms**, not a principal-agent relationship.

11-C:

To internalize a **negative externality in consumption**, such as smoking or sugary drinks, the most efficient market-based solution is to impose an **indirect tax** on consumers. This tax raises the private cost of consuming the good by adjusting them for external costs, reducing quantity demanded and ideally moving equilibrium closer to the **socially efficient output**, where $MSB = MSC$. By adjusting prices to reflect external costs, the market outcome becomes more socially optimal. This is why, option C, becomes correct.

A: Information provision (education campaigns) is useful, but it’s not a strong market-based mechanism; **it directly targets awareness**, not price. Also, the advertisement may be ineffective in reducing demand. Most importantly, the impact of advertisement to nudge individuals and influence their behavior is observed in the long-term, thus making it incorrect as it is not the most-effective market-based solution.

B: Subsidizing producers would **lower** prices and increase consumption — the opposite of what is needed for a negative externality.

D: Doing nothing ignores the market failure — free markets under- or over-produce when externalities exist.

12-D: The justification for government replacing a private firm in collecting waste may lie in the **profit motive** because a private firm, **primarily operating with the objective of “profit-maximization” might put profits before consumer needs**. As a result, it may prioritize **cost-cutting**, which may potentially lead to poor service or insufficient concern for **externalities** (like pollution). On the contrary, a state-run firm can prioritize public welfare.

A: Waste collection is **not a public good**. It is **excludable** and **rivalrous** — public goods must be both non-excludable and non-rival.

B: Government funding doesn’t guarantee lower costs — there is a greater risk of public sector firms being inefficient due to lack of competition.

C: The claim that private firms are “always less efficient” is incorrect — many are more efficient due to competition.

13-D:

What is a government failure?

Government failure occurs when government intervention in the economy, intended to address market failures or improve overall outcomes, instead creates inefficiencies, exacerbates existing problems, or leads to unintended negative consequences

→ **Essentially, it's when government action makes things worse than they would have been without intervention.**

D) Tariffs are intended to protect domestic producers, but they can also raise prices of imported goods for consumers and reduce economic welfare. If these tariffs **worsen income inequality**, as shown by an increasing **Gini coefficient**, then the intervention has produced **unintended adverse consequences** — a classic case of government failure. This is why, key D, becomes correct.

A: This is an example of successful intervention — internalizing external costs.

B: Improving labor mobility by reducing union restrictions promotes efficiency.

C: Removing price controls helps restore free-market signals — this does not cause a government failure

14-C:

The **marginal revenue product (MRP)** of labor (**or the demand for labor**) is the product of a worker's marginal product of labor (MPL) and the revenue earned upon selling the additional output (MR). So, if the **price of electric vehicles rises**, this will encourage producers to extend their quantity supplied of electric vehicles, representing an increase in production. When more output needs to be produced, more labor will have to be hired to increase the production- since the **demand for labor is derived demand**.

This rise in demand for labor means that the DL curve or the Marginal Revenue Productivity Curve will shift to the right. This is why, key C, becomes correct.

A: A fall in the price of petrol vehicles will extend QD of Petrol Vehicles. And it will, on the other hand, decrease demand for electric vehicles, which is a substitute. In response to the decreased demand of EVs, firms will reduce their demand for labor producing EVs, thereby causing their MRP curve to shift to the left.

B: Lower productivity would reduce MPL and subsequently MRP as well, shifting the MRP curve to the **left**, not right.

D: Wage rate changes bring a movement in the demand for labor curve; they won't change the position of the curve.

15-C:

$MRP = MPL \times MR$ (which is equal to the price, as in perfect competition, $AR=MR$)

Therefore,

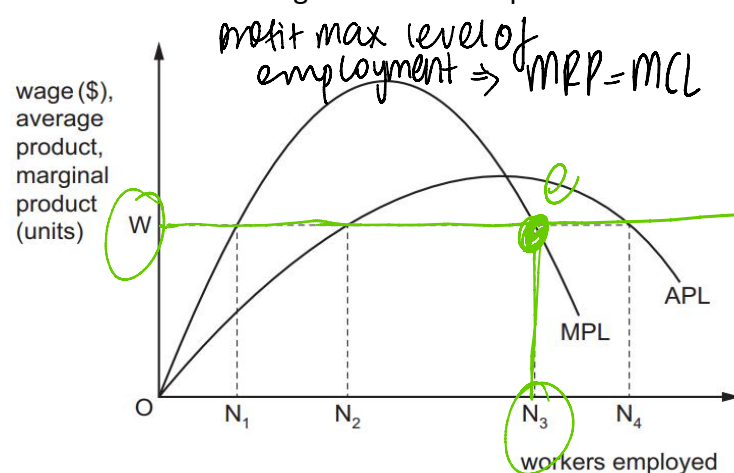
$MRP = MPL \times P$

In perfect competition (product market), **P is constant**, because the firm is a price taker.

Important Note: So while $MRP \neq MPL$ in absolute value, the shape of the MRP curve is similar to the MPL curve, just scaled up or down by the constant price.

Since we don't have MRP data, we will utilize the MPL curve to decide the profit-maximizing level of employment.

- A profit-maximising firm will hire labor until the marginal revenue product (MRP) equals the wage.
- But since, we are using the data of MPL as mentioned above, this happens at ON_3 workers — MPL intersects wage line W at this point.



$SL = APL = MCL$

* In a perfectly competitive labor market, the shape of the MPL curve and the MRP are 'similar' since MRP is MPL multiplied a constant price.

16-B:

- A) A **decrease** in credit card usage means people rely **more** on cash → demand for money **rises**, not falls.
- B) Switching to weekly wage payments reduces the average amount of money people need to hold for transactions, as income is received more frequently. This lowers the demand for money.
- C) A fall in interest rates would bring about a decrease in savings by reducing the incentives to save, and it would consequently increase spending. Hence, it will raise the transactionary demand for money.
- D) More risk in holding bonds means that people prefer to hold **money**, so money demand **rises**.

17-C:

Structural Unemployment includes:

- ➔ **Technological unemployment**
- ➔ **Regional unemployment**
- ➔ **International unemployment**

A) An increase in lending by low income households would increase consumption by them, which will result in an increase in Aggregate demand, thereby resulting in a decrease in demand-deficient unemployment. Here we are interested in an option that represents a reduction in structural unemployment.

B) If subsidies are introduced for technological improvements, this will translate into an increase in technological unemployment, which is a part of structural unemployment.

C) An increase in tariffs makes imported goods more expensive, encouraging consumers to buy local products. This boosts domestic industries, leading to more job creation. **It will reduce international unemployment, which means that structural unemployment will reduce.** At the same time, lower taxes for new businesses reduce their costs, making it easier for them to grow and hire workers. New jobs being created can help reduce structural unemployment by aligning labor demand with available skills in the economy.

D) An introduction of National Minimum Wage makes hiring labor expensive, thereby resulting in a contraction in labor demand. Alongside which, quantity supplied of labor will extend. This will result in an increased unemployment.

18-A:

Natural Rate of Unemployment includes frictional, seasonal and structural unemployment.

A higher **natural rate of unemployment** may stem from **rigidities** in the labor market.

Strong **trade union power** can contribute by keeping wages above equilibrium, discouraging hiring and leading to **real-wage unemployment**. Additionally, a stronger trade union can also contribute to an inflexible workforce, thereby also contributing to an increased **natural rate of unemployment in country X**

B: More training and education in X would make workers more versatile and multi-skilled, and a rise in occupational immobility would reduce structural unemployment, leading to a **lower** natural rate of unemployment in X.

C: Stronger incentives to find work would lower voluntary unemployment, hence decreasing Natural Rate of Unemployment in X.

D: Better job information also reduces frictional unemployment, hence decreasing Natural Rate of Unemployment in X.

19-A:

To calculate real income per capita, we will need to adjust nominal income with inflation and the size of the population.

Real income growth = Nominal income growth – Inflation
= 2% – 3% = **–1% real income**

Per capita income growth = real income – population growth
= –1% – 1% = **–2%**

Thus, real income **per head** has fallen by 2%.

20-D:

Inflation due to rising **import prices** (cost-push) can be tackled by **revaluation** of the currency (which includes raising the exchange rate)

Since revaluation would make imports cheaper, this directly reduces imported inflation. This is because a stronger exchange rate lowers the **domestic price of imports**, easing inflationary pressure.

A: Lower interest rates may lead to a fall in the exchange rate (the exchange rate to depreciate), which can further raise the domestic prices of imported goods.

B: Lower income tax boosts disposable income, consumption and Aggregate demand → demand-pull inflation risk.

C: Tariffs would further raise import prices since they are a tax on imported goods, worsening cost-push inflation

21-C:

A closed economy is characterized by its isolation from the rest of the world, meaning it does not engage in trade or financial transactions with other countries.

In a **closed economy**, international trade is absent — meaning no exports, imports, and due to detachment from the rest of the world, foreign financial flows are also not present. Therefore, achieving a **surplus on the balance of payments** is not a meaningful objective, as the current account, capital account, and financial account all depend on international transactions. This is why, key C, is correct.

Other objectives like inflation, income equality, and growth remain relevant in a closed economy. A government may still pursue **low inflation, sustainable growth** through investment and productivity gains, and **more equal distribution** through taxes, benefits, and education.

22-C:

A government increases its **budget deficit** to invest in infrastructure and **finances it by printing money** (monetary expansion).

A budget deficit represents the usage of an expansionary fiscal policy, whereas monetary expansion represents an expansionary monetary policy. Both these policies would stimulate economic growth and lower unemployment, while increasing the risk of demand-pull inflation.

- This policy boosts **government expenditure (G)** → increases **aggregate demand (AD)** → leads to **higher economic growth**.
- The rise in AD reduces **cyclical unemployment**, since firms need more workers to meet rising demand.
- However, when a rise in AD outpaces a rise in the productive potential of the economy, there is a risk of **too much money chasing too few goods**, causing demand-pull inflation.

So, while growth and employment rise, **inflation also rises**, making **C** correct.

23-C: Raising interest rates is a classic **contractionary monetary policy** used to reduce inflation.

Higher interest rates increase the cost of borrowing and raise the return on saving. As a result, **consumption and investment** fall. This lowers **aggregate demand, reducing demand-pull inflation**.

However, reduced Aggregate demand would reduce actual economic growth, and may increase **unemployment**, particularly if the economy is already weak. Thus, one trade-off of using interest rates to control inflation is **lower economic growth**.

A: Higher interest rates attract foreign direct investment and hot money inflows, **appreciating** the currency, not depreciating it.

B: Higher interest rates result in an increased incentive to save, which will encourage an increase in savings, rather than reducing it.

D: Higher interest rates reduce AD, which can raise demand-deficient unemployment, not reduce it.

24-C:

The diagram represents a **Laffer Curve**, which shows that tax revenue initially rises as tax rates increase (**representing a positive relationship between tax rate and tax revenue**)—but beyond a certain point (**Z**), higher tax rates discourage work, and encourage an increase in tax evasion, causing total tax revenue to fall (**representing a negative relationship between tax rate and tax revenue**)

Past point Z, increasing tax rates creates **disincentives to work more, and people are more likely to avoid tax by finding loopholes**. Thus, **the greater the rate beyond Z, the smaller the revenue generated**, making C correct. This is why, key C, is correct.

A: The cut from Y to Z would cause the tax revenue to increase.

B: At tax rates below Z, a cut in tax-rate would cause revenues to fall.

D: Revenue doesn't **always** rise with tax rates — that's the essence of the Laffer Curve.

25-C:

Devaluation of a currency raises the **domestic price of imports**, including raw materials:

If **local substitutes are unavailable**, this means that **the demand of imported goods is price-inelastic**, and firms **must import** at higher prices. This rise in import expenditure increases **costs of production**, shifting **aggregate supply leftward**, causing **cost-push inflation**.

So, lack of domestic substitutes for imported goods makes an economy **highly vulnerable** to import-cost-driven inflation.

A: Devaluation results in a BOT Surplus in the long-run by making export oriented goods cheaper and imported goods expensive, which results in a rise in Aggregate demand. However, excess capacity in the economy would allow the economy to grow without the risk of demand-pull inflation.

B: Reducing tariffs makes imports cheaper — the opposite of inflationary.

D: Inelastic export demand means that even after a decrease in export prices, export qty demanded wouldn't extend by a higher percentage, and export revenue will fall. It will reduce injections in the circular flow, and bring about a decrease in Aggregate demand. As a result of this, demand pull inflation would reduce.

26-A:

If a country imports most of its factor inputs, then imported inflation is a risk.

An **appreciation of the exchange rate** reduces the domestic price of imports. This lowers **production costs**, shifts **AS rightward**, and helps control **cost-push inflation**. This is why, key A, becomes correct.

B: Income tax increases will reduce **AD** (demand-pull inflation), not cost-push inflation.

C: Higher interest rates reduce **AD** (demand-pull inflation) but may not directly ease imported cost pressure.

D: Sales tax is a tax on goods and services. If a tariff is imposed on imported goods, it will further raise the prices of imported goods, exacerbating cost-push inflationary pressures.

27-D:

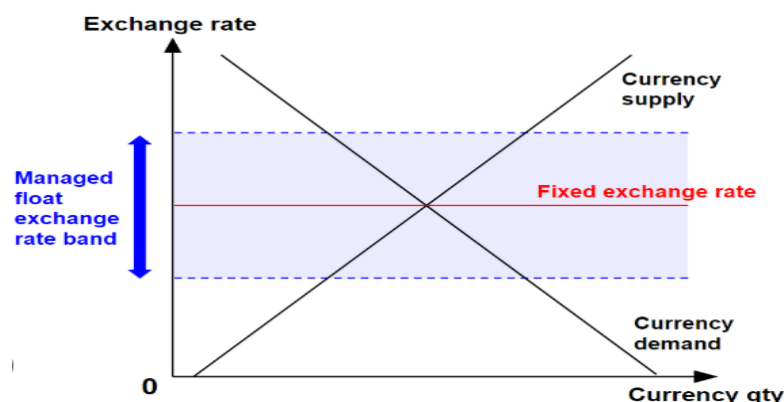
The **current account** includes:

- ➔ Balance of Trade in goods and services
- ➔ Primary Income (profits, wages, interest-income from abroad)
- ➔ Secondary Income (e.g. foreign aid sent or received/remittances)

Capital and financial transactions, such as **foreign direct investment (FDI)** like investment in pipelines, are part of the **financial account**.

So, D is excluded from current account calculations.

28-C:



A **managed float** or **dirty float** means the exchange rate is mostly market-determined but with occasional **central bank intervention** to keep it within a desired band. The country **allows some fluctuation**, but steps in to stabilize when necessary.

A: A fixed exchange rate doesn't allow any movement.

B: A floating rate allows full movement without intervention.

D: A trade-weighted index is not a type of exchange-rate regime.

29-A:

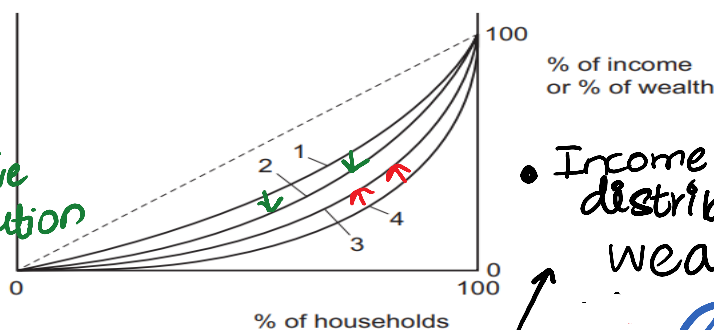
A – Income: curve 1 → 2, Wealth: curve 4 → 3

Lorenz curves graphically represent the distribution of income or wealth. The further the curve lies from the 45-degree equality line, the **greater the inequality**. In this question, income inequality is increasing (more unequal), and wealth inequality is decreasing (more equal).

Curve 1 is closer to the line of perfect equality than curve 2, so a shift from 1 to 2 reflects a **worsening income distribution**.

Curve 4 is farther from the line than curve 3, so moving from 4 to 3 indicates **improved equality in wealth distribution**. Therefore, A accurately captures both movements.

29 The Lorenz curves in the diagram show different distributions of income and of wealth.



• Income = more equally distributed than wealth.

4-3

At first, income in a country is more equally distributed than wealth.

In a period, the distribution of income becomes more unequal but the distribution of wealth becomes more equal.

Which movement would show the effects of these changes on the distribution of income and wealth within the country?

	Income Inequality ↑ distribution of income	Wealth Inequality ↓ distribution of wealth
✓ A	shift from <u>curve 1</u> to <u>curve 2</u>	shift from <u>curve 4</u> to <u>curve 3</u>
B	shift from curve 2 to curve 1	shift from curve 3 to curve 4
C	shift from curve 3 to curve 4	shift from curve 2 to curve 1
D	shift from curve 4 to curve 3	shift from curve 1 to curve 2

1-2:
*Initial income Lorenz curve (1) = more equal distribution than wealth

* 1-2 (Lorenz curve moves further away from the line of perfect equality)

* 4: Initial wealth Lorenz curve (4) = less equal distribution than income

* 4-3: Lorenz curve moves closer to line of perfect equality.

30-B:

B: As countries develop, improvements in healthcare, sanitation, education, and income levels lead to better living conditions and medical care. This results in a higher average life expectancy, a core indicator of human development.

A: With development, access to higher education expands, and enrolment rates typically increase.

C: Development leads to lower birth rates due to better education and healthcare, especially for women, so population growth rates generally decline.

D: Incorrect. In more developed economies, the tertiary (service) sector expands as demand shifts from goods to services like health, finance, and education.

